HUNGARIAN CENTRAL STATISTICAL OFFICE

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New data sources and AI

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Web Intelligence Challenge

EUROPEAN STATISTICS AWARDS
WEB INTELLIGENCE

- **The task:** classifying occupations (ISCO) for online job advertisements
- **Online job advertisements contain various types of information** including a job description, information about the company looking to hire, job benefits, requirements for job seekers, etc.
- In order to **calculate meaningful statistics** given the data collection method and size of the online job advertisements datasets, **occupational class labels must be provided** for these various entries.
- Within the WI CLASSIFICATION CHALLENGE, teams will compete **using advanced modelling techniques to develop an efficient and robust automated solution for correctly assigning class labels**.

Web Intelligence Challenge

Provided were:

- The ISCO (International Standard Classification of Occupations) nomenclature
- A dataset of ≈26 000 online job advertisements
- **Open source tools**

0110,Commissioned armed forces officers,"Commissioned armed forces officers provide leadership and management to organizational units in the armed forces and/or perform similar tasks to those performed in a variety of civilian occupations outside the armed forces. This group includes all members of the armed forces holding the rank of second lieutenant (or equivalent) or higher.

```
id,title,description
872828466,Panel & Paint Technician,"Panel & Paint Technician required in Colchester, Essex Hours: Monday to Friday â€“ 45 hours 7.00am-4.30pm Basic Salary: Up to ÂŁ35,000
839465958,"LĂ¤rare i slĂ¶jd och teknik fĂ¶r ĂĄrkurs 7-9, Ljungsbro skola","Sista ansĂ¶kningsdatum: 1 juni 2021 Referensnummer: 7040 AnstĂ¤llningsform: TillsvidareanstĂ¤l
857077872,Consultants in Emergency Medicine - Doughiska,"The Galway Clinic is a leading 146 bed, state of the art independent hospital situated on the outskirts of Galway, I
801801567,Senior IT Support Engineers,"My Client, who has been continually growing throughout the pandemic, is currently recruiting for 2 Senior IT Support Engineers to joi
```

Aims

- Explore the open source possibilities
- Build skills
- Gain hands-on experience
- Create a multi-purpose toolbox
- Make plans to produce experimental statistics

Problem Setting and Preparation

- **Classification**: ISCO-08 has 436 categories at the fourth level
- **Unsupervised**: there are no existing labelled data!
- **Preparation:**
  - ✓ python environment set up
  - ✓ data import
    - ❑ UTF-8 input csv → pandas df
  - ✓ multilingual translation
    - ❑ deep-translate and langdetect library → Google Translate API
    - ❑ ~ 4 hours running time for about 5000 non-English texts
    - ❑ ~ 1,5 hours with dask df
  - ✓ text pre-processing
    - ❑ removing nouns, lemmatising with WordNet etc.

Early Approaches – TF-IDF models

- Term frequency–inverse document frequency, is **a measure of importance of a word** to a document in a collection or corpus, adjusted for the fact that some words appear more frequently in general
- **Vectorisation** of the texts
- Creation of a **vocabulary**, each word = a one-hot vector
- **Embedding** the job advertisements and the ISCO descriptions
  - Similarity: cosine of vectors (dot product)
  - Majority approach: the most similar category starts with ”2”, but the other four in the top five most similar starts with ”3”, then forced to start with ”3”.
  - Hierarchical approach: if we already decided that a code starts with ”31”, then we choose from the most similar subcategories of 311, 312, 313, 314 and 315.

Early Approaches – Fine-tuning TF-IDFs

- The vocabulary was fine-tuned by setting the **”drop most common”** parameter, i.e. removing words that are too common, hence don’t serve as useful information.
- Improvement with the use of 2-grams (instead of ”software” and ”engineer” we have ”software engineer”)
- **”Mixing”** parameter to set the amount of 2-grams to be used (embeddings and calculations with n-grams are computation-heavy)

New approach – use of LLMs

- The problem with the common techniques was that **job advertisements and ISCO descriptions are very different** in style, structure, wording etc.
- The idea was to use LLMs to **generate synthetic data** (OpenAI API)
    - ChatGPT-4o generated job ads based on ISCO descripitons weren’t diverse enough
    - Rather use the job examples in the ISCO descriptions
    - Experimenting with prompting and temperature settings [$$]
    - Batch calls: 325 token texts × 14 examples × 436 categories = 2M tokens ≈ 10$

New approaches – Upgraded TF-IDFs

- TF-IDF + logistic regression
- TF-IDF + Naive Bayes

Training set	Model	Result
ISCO descr.	TF-IDF	23%
ISCO descr. + Chat GPT	TF-IDF + logistic regression	36%
ISCO descr. + Chat GPT	TF-IDF + Naive Bayes	40%

```
# Label encoding
label_encoder = LabelEncoder()
df['encoded_codes'] = label_encoder.fit_transform(df['codes'].tolist())

train_texts = df['description'].tolist()
train_labels =  df['encoded_codes'].tolist()


# Use TF-IDF to convert text to features
tfidf = TfidfVectorizer(max_features=40000,
                        ngram_range=(1, 2),
                        stop_words=stopwords.words('english'))
train_tfidf = tfidf.fit_transform(train_texts)
test_tfidf = tfidf.transform(df_test['description'].tolist())

# Train a Naive Bayes model
model = MultinomialNB(alpha=0.1)
model.fit(train_tfidf, train_labels)

# Make predictions
pred_labels = model.predict(test_tfidf)
```

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Final solution – Use of Transformers

Training

- We used the transformers and wandb libraries to access pretrained models,
- i.e. trained LLM embeddings such as DistilBERT and RoBERTa

  (the input contained too little information to train a more complex model such as Bert*),
- then fine-tuned a 'DistilBERT-base-uncased' model with 436 output labels
- on the chatGPT generated data (14 example / ISCO code) and
- on the ISCO code's label itself.

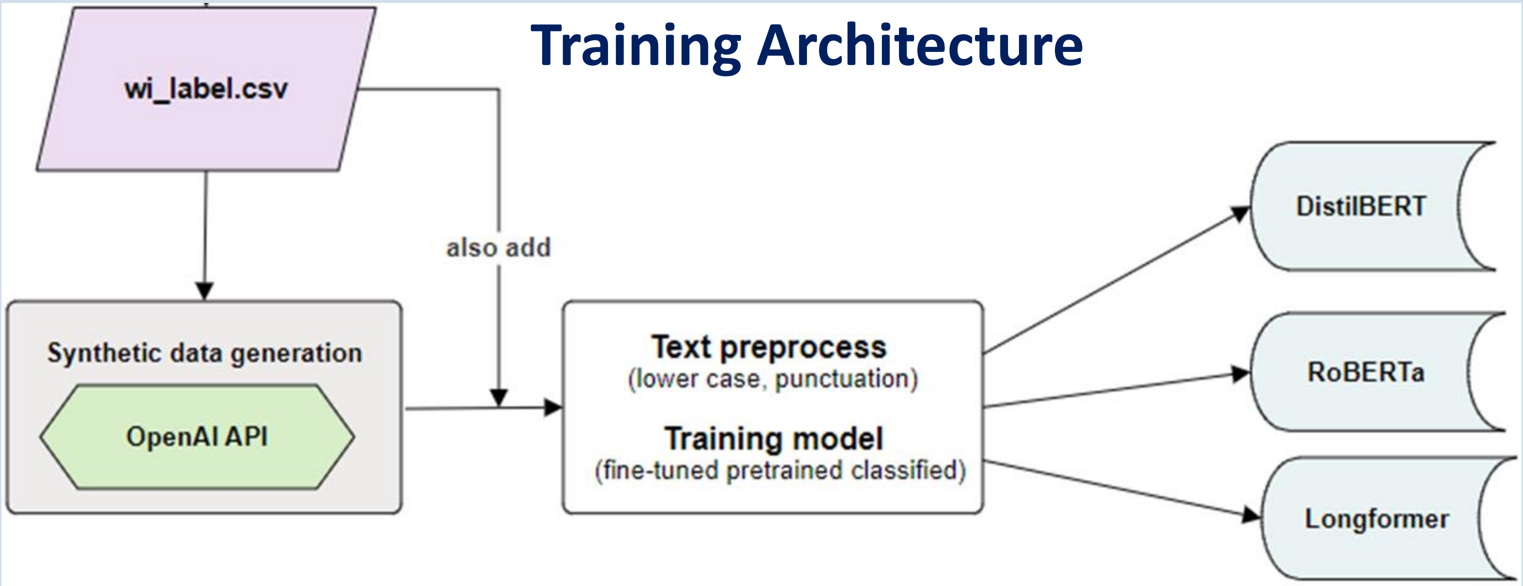
  ! rule of thumb: 50-100 sample/output label recommended,

  ! we had 15 to keep it open source and low budget (10$)

** BERT stands for* ***Bidirectional Encoder Representations from Transformers.*** *It is based on transformers, being a* ***deep learning model*** *in which every output element is connected to every input element, and the weightings between them are dynamically calculated based upon their connection.*

Final solution – Use of Transformers

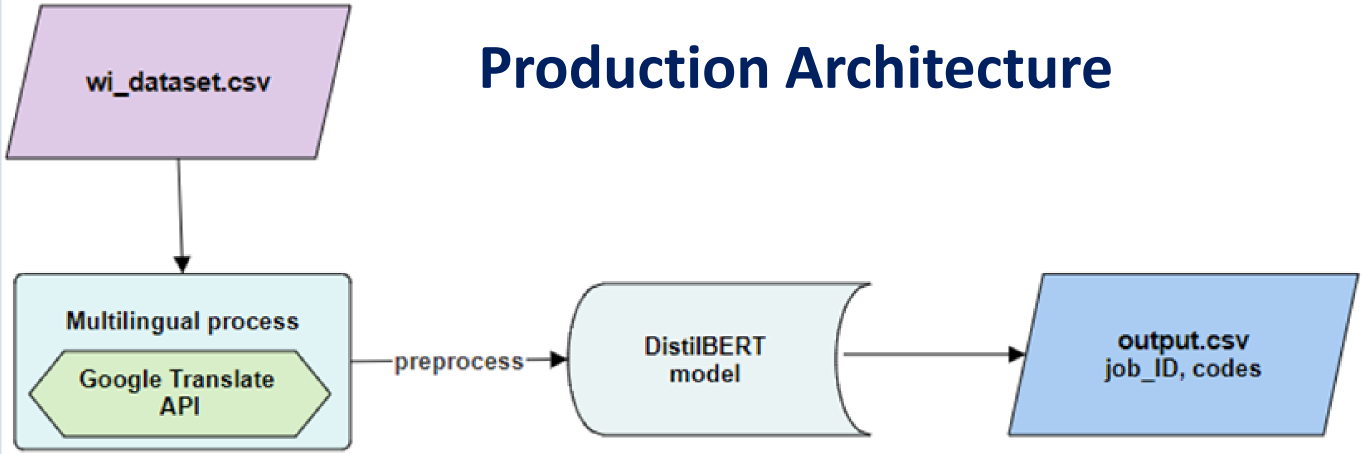
- The training not **only fine-tuned the embedding**,
- but also **classifying dense layers** were put on top of it,
- this way we got a **neural-network "regressor"**.



We trained for **4 epochs**.
The compiler was used with standard Adam (Adaptive moment estimation) optimizer and with CrossEntropy as the loss function.
Texts were loaded in batches of 8.
The training was carried out via Kaggle, using GPU P100 and took about **15-20 minutes**.

Final solution – Use of Transformers

- The input data is **transform**ed into a pandas df, then
- non-English ads are **translate**d and
- after minimal **preprocess**ing
- the ads are getting ISCO **label**led in the output file.



- This model starts from labelling the ads correctly at 50% on the first digits and 35% on the full ISCO codes!
- This is where the non-transformer models' capabilities ended.

Evaluation of results

➢ How to think about the average accuracy of 40%?

- First of all ~ 40% accuracy is not a bad performance regarding automatic coding
- There are no existing experience in this particular field! (only official statistics is interested)
- There are no validated data to train on! Had to start from scratch.
- Kept it open source! (Apache 2.0 license)

Lessons learned and next steps

➢ What's new?

- LLM embeddings
- AI generating synthetic data
- Automatised online translation

➢ What to expect?

- by getting some validated data (e.g. the 26 000 labelled job ads) this same model would probably perform above 70%
- there is an ongoing Grant call
- produce experimental statistics about job vacancies
- use of these solutions in other areas of automatic coding

➢ Limitations

- data protection and legal regulations
- IT security
- financial burdens
- ethical concerns

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Thank you for your attention!

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